

Escape Room

Secret Chamber

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Project Overview

Games Structure

Rooms and Objects	Dictionaries of all doors, keys, objects and rooms
Object Relations	Connects all interactables with each other
Player Progress	Dictionary that tracks all player, location, puzzle and timer status
Replayability	Resets all object relations, inventory and images when starting a new game
Presentation	Added images for storytelling and different text colors for errors, success, alerts and narrator
Game Loop	Input the player decisions and check if time is running out
Mechanics	Checking if player has the necessary items to progress in the game

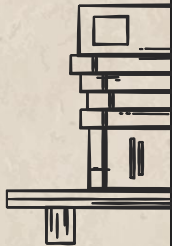
Functions

- Added time library (`import time`) and created a function to calculate how much time passed since the player woke up, the game state `start_time` value is `None` that will keep adding up with `time.time()`, if time runs out is "Game Over".
- Added copy library (`import copy`) so that Python creates a copy of the object relations dictionary without affecting the original, this makes the game replayable and resets the objects into their places automatically when `start_game()` is initialized.
- Added (`import display, Image`) to render and display visual images directly in the output, making the game more interactive.



Features and Additions

- **Changed "Explore" and "Examine" typing requirement to "1" and "2"**
- **Added a Countdown Timer to the Player to track time during the game by typing "3"**
- **Added Display feature for Room Images and preventing them from appearing more than once in the same game**
- **Added Display feature for Room Descriptions and preventing them from appearing more than once in the same game**
- **Added Colors to text distinguishing between actions, descriptions, Items and alerts.**
- **Added 4 Room Type Dictionaries** ("Kitchen", "Storage Room", "Stairs" and "Basement")
- **Added 4 Key Type Dictionaries** ("Key E", 2x "AA+ Battery" and Hammer)
- **Added 3 Door Type Dictionaries** ("Door E", "Door F" and "Window")
- **Added 1 Lock Type Dictionary** ("Door G")
- **Added 10 Furniture Type Dictionaries** ("Fridge", "Kitchen Cabinet", "Kitchen Sink", "Trash Can", "Bedside Table", "Sofa", "Coffee Table", "Shelf", "Toolbox" and "Flashlight")
- **Added 8 Images for Room Type Dictionaries** ("Game Room", "Bedroom 1", "Bedroom 2", "Living Room", "Kitchen", "Storage Room", "Basement", "Outside")
- **Added 1 Image for Game Over when Time Runs Out**
- **Added a Combine feature** (Batteries and Flashlight)
- **Added a Stairs condition locked until player finds Items** (Batteries and Flashlight)
- **Added a code to user** Input on Door G



Technical Challenges

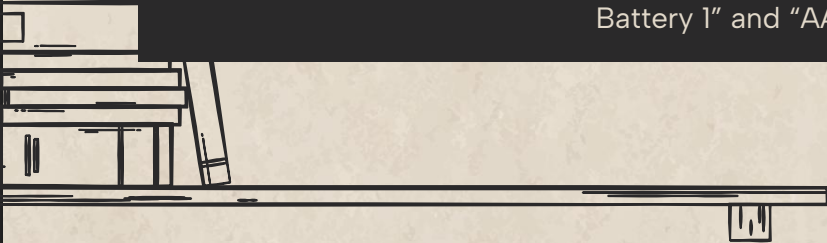
The 2 most technical challenges we faced during this project:

First was the order of the conditions inside the `examine_item()` function. The condition that unlocks door F and door G must come on top of the condition that unlocks all doors, otherwise it will never reach them and the player keeps locked out.

Second, the creation of the condition for the flashlight to combine with the 2 +AA Batteries and check if the player can go down the stairs or not.

To overcome the first challenge we kept testing and moving the order of the conditions and asking AI to explain us more in depth the logic inside them.

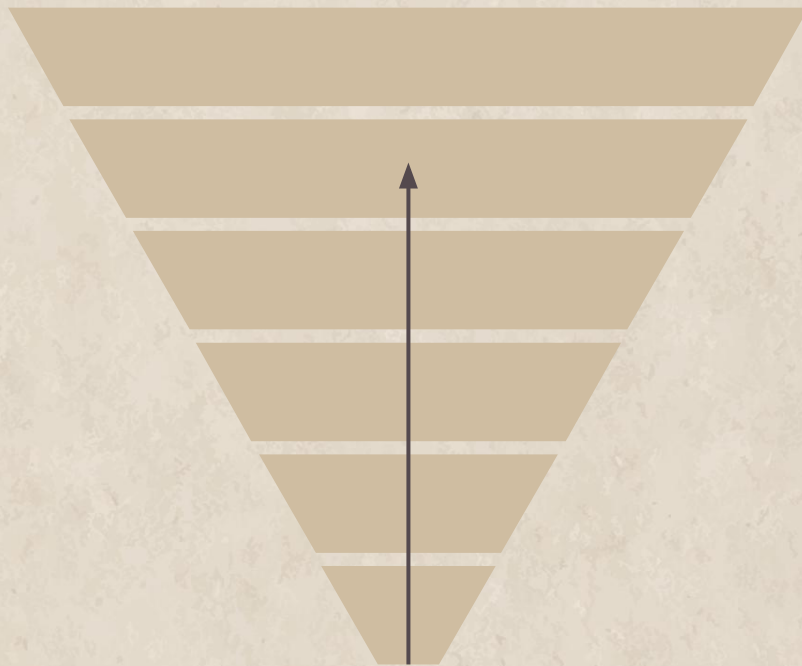
For the second challenge we took a closer look at how the function that unlocks all doors works but instead of looking for a single type key we wrote it so that the player must have all three type key items ("Flashlight", "AA Battery 1" and "AA Battery 2") on their inventory.



Biggest Mistake



Our biggest mistake was to think in very small details at the beginning of the project without first having a solid base of code and ideas. We think this took more time than expected and would be beneficial if we started from the foundation first.



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Secret Chamber Map



Demo

Thank You!

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